CLOUD COMPUTING FUNDAMENTALS

Beginner-Friendly Study Material

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11. Introduction to Cloud Computing

What is cloud computing?

I壱Definition

Cloud computing is a way to use computing services—such as servers, storage, databases, networking, and software—over the internet instead of owning and managing all the hardware yourself.

I壱Simple example

When you save photos in Google Drive or use an online document editor, your files or the application may be stored and processed on computers in a data centre. You access the service through the internet.

I壱Why it matters

Cloud services let people and organisations access computing resources when they need them. They can often increase or decrease their usage without buying new physical equipment.

I壱Key terms

A data centre is a facility containing computing equipment. A server is a computer that provides services to other computers. Storage keeps data, while a network connects devices and services.

I壱Key points

On-demand access: request resources when needed.

Internet access: use services from supported devices and locations.

Shared infrastructure: a provider may serve many customers while keeping their data and access separated.

Measured usage: some services charge according to how much is used.

Key takeaway: Cloud computing is not simply “saving files online.” It can provide computing power, databases, software, networking, analytics, and many other services.

12. Essential Characteristics and Benefits

How cloud computing helps

I壱On-demand self-service

Users can provision supported resources—such as storage or virtual machines—without waiting for a provider employee to configure each request.

I壱Broad network access

Services are available over a network and can be used from different types of devices, depending on the service and its security settings.

I壱Resource pooling

A provider pools computing resources to serve multiple customers. Resources are allocated dynamically, with logical separation between customers.

I壱Rapid elasticity

Capacity can expand or shrink when demand changes. For example, an online store may need more computing capacity during a sale.

I壱Measured service

Usage can be monitored, measured, and billed according to the provider's pricing model.

I壱Key points

Cost flexibility: reduce the need to buy and maintain all infrastructure upfront.

Scalability: adjust resources as needs change.

Speed: create many resources quickly.

Accessibility: reach services from different locations with internet access.

Maintenance support: the provider manages some underlying infrastructure, depending on the service model.

Key takeaway: Cloud can improve flexibility and speed, but it does not automatically make every system cheaper, safer, or faster. Good design and monitoring still matter.

13. Cloud Service Models

IaaS, PaaS, and SaaS

I壱Infrastructure as a Service (IaaS)

Provides basic computing resources such as virtual machines, storage, and networking. The customer has more control over the operating system and applications, while the provider manages the physical infrastructure.

I壱Platform as a Service (PaaS)

Provides a managed environment for building, testing, and deploying applications. Developers can focus more on their code because the provider manages more of the platform.

I壱Software as a Service (SaaS)

Provides a complete application over the internet. Users usually access it through a browser or app, while the provider manages most of the underlying technology.

ModelThink of it asExample useIaaSRenting computing building blocksConfigure a virtual server for a projectPaaSUsing a managed app-building platformDeploy a web app without managing the server OSSaaSUsing ready-made softwareEmail, online documents, or web-based collaboration**Key takeaway:** A simple memory aid: IaaS gives infrastructure, PaaS gives a platform to build on, and SaaS gives software ready to use. Exact responsibilities vary by provider and service.

14. Cloud Deployment Models

Where and how cloud resources are used

Public cloud

Cloud resources are provided by a third-party provider and made available to customers over a network. Customers share the provider's overall infrastructure, with isolation controls between their environments.

Private cloud

Cloud infrastructure is dedicated to one organisation. It may be managed by that organisation or a service provider and may be located on-site or elsewhere.

Hybrid cloud

Combines private infrastructure or a private cloud with public cloud services, allowing data or applications to work across both environments when designed to do so.

Community cloud

Infrastructure is shared by organisations with common needs, such as security, policy, or regulatory requirements.

Key points

Public cloud: useful when an organisation wants flexible resources without owning all the hardware.

Private cloud: may suit organisations needing dedicated control, though it can require more management and cost.

Hybrid cloud: can keep some workloads in one environment while using another for additional capacity or services.

Community cloud: can spread infrastructure costs across organisations with similar requirements.

Key takeaway: There is no single best deployment model for every situation. The choice depends on security, compliance, cost, control, skills, and technical needs.

15. Core Cloud Components and Common Uses

What cloud platforms provide

Compute

Compute services run programs and applications. Examples include virtual machines, containers, and serverless functions.

Storage

Storage services keep files, backups, application data, and other digital information. Different storage types are designed for different access patterns.

Databases

Cloud databases store and organise structured or unstructured data. Managed database services can handle tasks such as updates and backups, depending on the service.

Networking

Networking connects cloud resources and controls how they communicate. Virtual networks, routing, firewalls, and load balancers are common building blocks.

Identity and access management

Identity and access tools control who can sign in and what actions they are allowed to perform.

Monitoring and backup

Monitoring helps teams observe performance and errors. Backups help recover data after accidental deletion, failure, or other incidents.

Key points

Education: host learning portals and share course materials.

Business: run websites, customer applications, and collaboration tools.

AI and data: store datasets, run analysis, and access computing capacity for machine-learning tasks.

Development: create test environments without purchasing a physical server.

Key takeaway: Most cloud solutions combine several components. A web application might use compute to run code, a database to store information, and networking to connect users.

16. Security, Challenges, and Best Practices

Using the cloud responsibly

Shared responsibility

Cloud security is shared between the provider and the customer. The provider is responsible for certain parts of the service, while the customer remains responsible for tasks such as managing user access and protecting data. The exact split depends on the service model.

Identity and access

Give each user only the permissions needed for their role. Use multi-factor authentication where available and remove access that is no longer required.

Data protection

Protect sensitive information with suitable access controls and encryption options. Keep backups and test whether data can actually be restored.

Misconfiguration

Incorrect settings—such as exposing a storage resource publicly by mistake—can create risks. Review permissions and configuration before deployment.

Cost management

Unneeded resources can continue to incur charges. Monitor usage, set budgets or alerts where available, and remove resources that are no longer needed.

Availability and dependency

A service may be affected by network problems, provider outages, or dependence on a specific platform. Plan for important failures and understand recovery options.

Best practices

Use strong, unique passwords and multi-factor authentication.

Apply least privilege: grant only the access required.

Keep software and configurations updated.

Back up important data and test recovery.

Monitor usage, security alerts, and costs.

Do not upload private or sensitive information to a service unless you are authorised and understand its policies.

Key takeaway: Cloud services provide useful security tools, but secure results require correct configuration, responsible access management, and ongoing monitoring.

17. Example, Glossary, and Revision

Putting the concepts together

I壱Example: a student study website

Imagine a study website where students read notes and take quizzes. A compute service runs the website, storage keeps uploaded images and files, a database stores quiz results, networking connects users, and identity controls protect student accounts. Monitoring helps the developer notice errors, while backups help protect important data.

I壱How it might scale

If many students visit before an exam, the developer may increase the website's capacity or use automatic scaling if the chosen service supports it. After traffic drops, capacity can be reduced to avoid paying for unused resources.

I壱Important reminder

This is a simplified example. A real design must consider privacy, security, reliability, accessibility, performance, and cost.

I壱TermSimple meaningCloud computingUsing computing services over a network, often the internetData centreFacility that houses computing equipmentScalabilityAbility to handle growth or changing demandVirtual machineSoftware-based computer running on physical hardwareIaaSCloud infrastructure resourcesPaaSManaged platform for application developmentSaaSReady-to-use software serviceDeployment modelHow cloud infrastructure is organised and made availableMFAMulti-factor authentication; more than one proof of identityBackupCopy of data kept for recoveryQuick revision questions

1. What is cloud computing, in your own words?
2. Name the three main cloud service models.
3. How does IaaS differ from SaaS?
4. Give one example of a public-cloud use.
5. Why are access controls and backups important?

Key takeaway: Cloud computing gives flexible access to technology. Understanding service models, deployment choices, core components, security, and cost helps you make better decisions.